

TEST TASK: HANDS ON

POSITION:	AI Operator
Overview	Build a knowledge base system that ingests email threads exported from Gmail and makes them searchable through AI-powered question answering. The system should accept a natural-language question and return an accurate answer along with references to the source email(s). Time est.: up to 3 hours
Constraints	Language: Python, JavaScript, or Go All technology choices must be justified in your README: explain why you chose each tool, library, or service.
Provided	A sample Google Takeout export <code>sample-takeout.zip</code> containing approximately 50 email threads in standard <code>.mbox</code> format. This is the same format Google produces when you export your Gmail data via Google Takeout <code></code> https://support.google.com/accounts/answer/3024190?hl=en The <code>.mbox</code> file contains multi-message email threads with standard headers (<code>From</code> , <code>To</code> , <code>Cc</code> , <code>Date</code> , <code>Subject</code> , <code>Message-ID</code> , <code>In-Reply-To</code> , <code>References</code> , <code>X-GM-THRID</code>).

Requirements

1. Data Ingestion

Parse the provided Google Takeout `.mbox` file and extract email threads with full context:

- ☐ Sender, recipients (To, Cc)
- ☐ Subject line
- ☐ Date and timestamps
- ☐ Message body (plain text)
- ☐ Thread relationships

2. Storage

Store the parsed email data in a way that supports retrieval for AI-powered Q&A. You choose the storage solution — it could be a vector database, a traditional database with full-text search, a hybrid approach, or something else entirely.

The key requirement: when the AI answers a question, the system must be able to identify and retrieve the relevant email thread(s).

3. AI-Powered Q&A

Connect an AI model so that a user can:

- ☐ Ask a natural-language question about the email data
- ☐ Receive an accurate answer
- ☐ See source references for the answer, specifically: which email thread, who sent it, and when

You design the AI approach. This could be RAG (retrieval-augmented generation), direct context injection, or any other method you find appropriate. You choose the AI provider (OpenAI, Anthropic, Google, open-source models, etc.).

4. Interface

Build a functional interface for interacting with the knowledge base. This can be: a web application, CLI interface, Slack bot, etc.

The interface must support:

- ☐ Asking a question in natural language
- ☐ Displaying the answer
- ☐ Showing source references (thread subject, sender, date) for each answer

5. Setup & Documentation

Your `README.md` must include:

Setup instructions (how to install dependencies and configure the project. One-command setup is ideal)

How to run (how to ingest the data and use the Q&A interface)

Technology choices (for each major component (e.g. storage, AI model, interface), explain why you chose it and what alternatives you considered)

Out of Scope

Automated tests
CI/CD pipeline

*The following are
explicitly **not required***

User authentication
Production deployment
Real Gmail API integration (work with the provided export only)

Submission

Share a link to a private GitHub repository containing your solution. The repository must include everything needed to run the project locally using your provided instructions.

Share repository with: `denys.rekun@partner-way.com`

Provided export:
sample-takeout.zip